PROJECT EXPO' 26

GESTURE CONTROLLED CAR WITH ROBOTIC ARM

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Introduction:

Traditional robotic control via joysticks is often non-intuitive and limits precision. This project develops a **Gesture-Controlled Robotic Vehicle with a 3-DOF Manipulator** to create a natural Human-Machine Interface. By fusing MPU6050 and Flex Sensor data, the system translates hand orientation and finger articulation into real-time wireless commands, enabling intuitive remote operations in complex or hazardous environments.

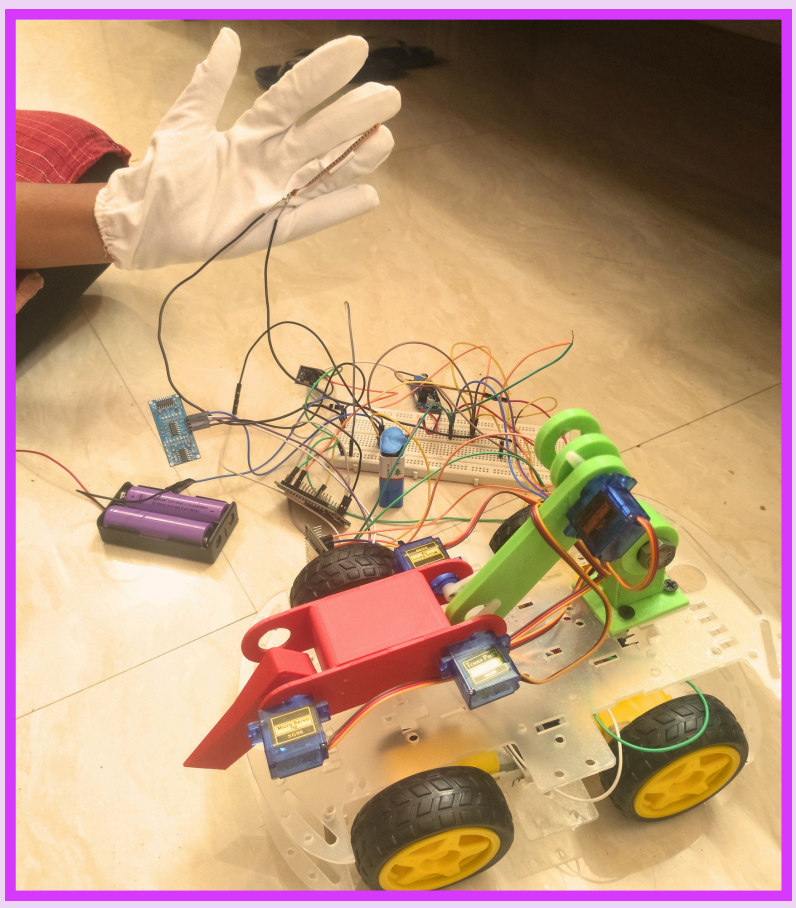
Methodology:

The system uses a transmitter-receiver architecture with two ESP32s communicating via the ESP-NOW protocol. The Control Glove (transmitter) processes MPU6050 tilt data for navigation and Flex Sensor input for gripper control into wireless commands. The Robot Chassis (receiver) translates these signals to drive an L298N motor driver and a 3-servo robotic arm, while an HC-SR04 Ultrasonic sensor provides an automated "Emergency Brake" if obstacles are detected within 20cm. Power is regulated through the L298N to provide a stable 5V rail for the microcontrollers and sensors, ensuring great performance.

Results And Analysis:

The gesture-controlled robotic system achieved high responsiveness and stability by successfully merging multi-modal sensor data. ESP-NOW proved superior to Bluetooth, delivering near-instant command execution with minimal latency. The MPU6050 provided precise tilt data for smooth navigation, while the Flex Sensor accurately controlled the gripper via calibrated analog thresholds.

Reliability was bolstered by the Ultrasonic sensor's "Collision Guard," which effectively overrode forward movement to prevent impacts. Despite the high current draw of the servos on the L298N regulator, sequential movement logic prevented system resets.



Conclusion:

This system demonstrates low-latency gesture control via ESP-NOW and sensor fusion. By integrating a 3D-printed 3-DOF arm with a mobile chassis, the project achieves precise, real-time manipulation, successfully bridging the gap between human intent and robotic action.

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